
SAFE on RoboCasa

Complete Experimental Report

Policy-internal failure prediction with π_0 , RLDX-1, Xiaomi-Robotics-1, and semantic Subtask-SAFE

Period: 13 July–15 August 2026
Report date: 15 August 2026
Primary benchmark: RoboCasa atomic and composite tasks
Primary detector families: independent MLP and temporal LSTM
Evidence base: checksum-verified archive plus frozen-holdout follow-ups

This report records the complete experimental chain: feature instrumentation, collection and export validation, π_0 pilots, RLDX scaling, natural-prevalence and loss-weighting controls, atomic/composite ablation, semantic Subtask-SAFE, causal within-horizon objectives, targeted collection, the frozen parent-disjoint evaluation, and the Xiaomi-Robotics-1 original-SAFE study over 39 tasks. Negative results are retained because they determine the next research direction.

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1 Technical summary

1.1 Question

Can SAFE-style detectors predict that a RoboCasa policy will fail early enough to support intervention, using internal action-policy features rather than only simulator state or elapsed time?

1.2 Answer

The integration and experimental protocol are valid, and RLDX-1 action features contain useful retrospective failure information. The strongest rollout-level RLDX MLP achieved task-normalized ROC-AUC **0.713 $\pm$ 0.017** on ten tasks with 25 successes and 25 failures per task. Atomic tasks were easier than composites. At conformal $\alpha = 0.20$, their balanced accuracies were **0.667 $\pm$ 0.030** and **0.590 $\pm$ 0.021**, respectively. Separate atomic-only and composite-only models did not improve matched performance, so mixed-task training was not the main problem.

Semantic Subtask-SAFE fixed the label mismatch for composites and achieved raw held-out segment ROC-AUC **0.734 $\pm$ 0.002**. However, this result did not survive the controls required for an online claim. A causal V2 pilot, a within-horizon target, temporal feature summaries, and 320 additional targeted parents were subsequently tested. The final experiment trained with 280 parents and preserved a frozen 20-parent calibration plus 80-parent evaluation allocation. At causal prefix 32, raw features achieved ROC-AUC **0.512 $\pm$ 0.003**; temporal features achieved **0.566 $\pm$ 0.004**. The parent-bootstrap intervals included chance, and neither detector passed the continuation gates.

The Xiaomi-Robotics-1 extension collected 1,950 new rollouts across 39 official mid-success tasks and pooled them with 692 earlier rollouts. A deterministic 1,950-rollout training export retained 38 trainable tasks after excluding one task with only one failure. The selected MLP and LSTM completed all 810 inner-CV fits and six final refits. Training-only task normalization raised MLP outer-test ROC-AUC from **0.532 $\pm$ 0.016** to **0.692 $\pm$ 0.013**. In causal replay, MLP SAFE showed early within-task signal, but its 13.2% FPR was not matched to time-only’s 5.3% FPR. The next confirmatory experiment is therefore a frozen, prospective 5%-FPR shadow evaluation on disjoint calibration and test collections.

Decision. RLDX Subtask-SAFE is useful as a retrospective diagnostic but is not yet a deployable early-warning signal. Xiaomi original SAFE is more promising: its normalized MLP contains early causal ranking signal, yet no matched-FPR prospective claim exists. Freeze the Xiaomi checkpoints, normalization, time baseline, stage windows, and 5%-FPR thresholds on a new calibration collection, then run a disjoint shadow test before any recovery intervention.

1.3 Key findings

1. ✓ The π_0 and RLDX feature-capture paths are official-loader compatible and store one feature per genuine policy inference.
2. ✓ Leakage-free inner cross-validation, fixed held-out evaluation, train-only normalization, and parent-grouped splitting are operational.
3. ✓ RLDX independent MLP features rank failures within tasks; additional data improved stability.
4. ✓ The normal RLDX alpha sweep reproduces the SAFE paper’s balanced-accuracy–detection-time evaluation geometry, with misses counted at normalized time 1.
5. ✗ The LSTM was consistently weaker than the independent MLP at these sample sizes.
6. ✗ Natural prevalence and official inverse-frequency weighting changed results only marginally.
7. ✗ Separate atomic/composite models did not beat the mixed model.

8. ✗ Raw Subtask-SAFE ROC was inflated by task/subtask score offsets, unequal priors, and late segment information.
9. ✗ SAFE did not beat task-conditioned elapsed time in causal-prefix or conformal operation.
10. ✗ Strong inner-CV results for within-horizon and temporal representations did not transfer to the frozen four-stage evaluation.
11. ✓ Xiaomi original SAFE completed a 39-task collection and a leakage-controlled 810-fit search; task normalization materially improved MLP ranking.
12. ✓ Xiaomi MLP SAFE showed task-macro causal ROC-AUC 0.671 at 10% and 0.626 at 25% of task horizon, above the within-task time-only value of 0.5.
13. ✗ The opened Xiaomi replay did not compare SAFE and time at the same false-positive budget; its promising early alarms therefore remain post-hoc.

2 Scope, labels, and experimental rules

2.1 Two prediction units

Rollout SAFE uses the official binary label: success if the RoboCasa terminal task predicate becomes true, otherwise failure at the horizon. **Subtask-SAFE** predicts whether the current ordered natural-language subtask eventually fails before completion.

One feature record is stored per genuine policy inference. Cached actions from a predicted chunk do not create duplicate feature records. This prevents action-reuse frequency from becoming an accidental sampling weight.

2.2 Models and selectors

The official SAFE detector families were retained:

- **indep**: an independent MLP detector;
- **lstm**: a temporal LSTM detector.

Hyperparameters include horizon selector, diffusion selector, learning rate, and regularization. Unless explicitly noted, the search used three inner folds and model seeds 0, 1, and 2.

2.3 Evaluation contract

- Outer test identities are fixed and never used for hyperparameter selection.
- Normalization statistics are fitted from training data only.
- Subtask segments are split by parent rollout before extraction.
- Raw pooled ROC/AP, task-normalized pooled ROC/AP, and macro task ROC/AP are kept distinct.
- Duration or elapsed-time baselines are mandatory.
- A conformal lead time is reported only together with the TPR that determines how often that lead is obtained.

3 Feature instrumentation and system validation

3.1 π_0 RoboCasa

The π_0 integration captures the official SAFE **pre_velocity** feature: action-expert suffix hidden states immediately before **action_out_proj**. The expected tensor is $10 \times 50 \times 1024$ (flow steps, action tokens, channels). Pinned instrumentation:

OpenPI commit: 5a6beda9ff99da30b4e1b59320f6a32971d7c397
 Patch: patches/openpi_safe_features_5a6beda.patch

The first smoke failed correctly because the policy server did not return `safe_features`. After applying the patch to the pinned server and enabling feature return, a sink smoke produced two successes and one failure and passed the official loader.

3.2 RLDX-1

RLDX has no literal `action_expert` module. The selected analogous action stream is `ao[:, -action_horizon:]` immediately before the embodiment-specific decoder. Its layer identifier is `action_model_msat_action_suffix_pre_action_decoder`; observed shape is $4 \times 16 \times 1024$.

Model: RLWRLD/RLDX-1-FT-RC365
 Repository commit: ef05cd4ae634ff97d672d42275febbc0b92cc192

Two ZeroMQ servers were validated on ports 20100 and 20101. A nested language-batch shape bug was corrected: flat request fields use `[instruction]`, while nested direct-policy language uses `[[instruction]]`.

3.3 Xiaomi-Robotics-1

The Xiaomi integration preserves the released RoboCasa365 policy behavior and adds an opt-in SAFE response. At each genuine inference it records the final DiT-layer action-token state immediately before the action output projection. The feature identifier is:

`dit_action_tokens_pre_action_output_layer`

Its observed tail shape is $5 \times 30 \times 1024$ (flow steps, action tokens, channels). One inference, not each cached action, creates one SAFE feature record. Pinned artifacts are:

Xiaomi repository commit: 4da1db0a4deefa6de7ebb4ef0b8754017290f5f7
 Checkpoint revision: 0d1aa76d0d82debc9b611e4d1e231096434d5be4
 Transformers: 4.57.1; action replanning interval: 16.

The official evaluation preprocessing was retained: `pretrain` split, observation history 4, interval 2, and crop ratio 0.95. The policy server and simulator client use separate processes and ports 10096 and 10097 for two-GPU collection.

3.4 ABot-M0.5

An ABot-M0.5 adapter was implemented with the same explicit request/response contract and inference-aligned capture rule. No ABot detector-training result is included here; it remains an engineering-only extension and must not be interpreted as evidence of model performance.

4 Dataset chronology and audits

Dataset/phase	Outcome	Evidence and interpretation
π_0 sink smoke	3 rollouts	2 successes, 1 failure; feature server and official loader validated.
π_0 first five-task attempt	invalid protocol	Wrong split/policy setup yielded only failures; discarded as scientific evidence.
π_0 five-task official	118 retained	60 successes, 58 failures; exact 10/10-per-task export derived from the healthy pool.
π_0 17-atomic expansion	incomplete	Asset and server failures preserved; not used in training claims.
RLDX 10-task pilot	200 retained	100 successes, 100 failures; 10/10 per task; five atomic and five composite tasks.
RLDX expanded	500 retained	250 successes, 250 failures; 25/25 per task; official export 15 GiB.
Natural-rate pool	689 outcomes	Genuine observed outcomes used to construct five immutable subsets with 23 training examples per task.
Subtask-SAFE parents	150 rollouts	65 successful, 85 failed parents; semantic decomposition into 18 task/subtask identities.
Subtask-SAFE segments	426 usable	341 successful, 85 failed; 27,420 policy records; official-loader compatible.
V2 targeted pilot	80 parents	Twenty new rollouts for each of four composite tasks; 36 successes, 44 failures; 257 semantic segments.
V2 long collection	320 parents	290-rollout long batch plus 30 additional Wash-Lettuce rollouts; all valid, with 202 failures and 118 successes.
V2 frozen allocation	100 parents	Twenty calibration parents and 80 evaluation parents, exactly 10 successful and 10 failed target-stage parents per stage; identities frozen before final refits.
V2 augmented training	280 parents	Original training parents plus 180 previously unassigned new-seed parents; frozen calibration and evaluation identities unchanged.
Xiaomi ten-task pool	692 rollouts	419 successes and 273 failures; deterministic 25/25-per-task export contained 500 rollouts.
Xiaomi official mid-success collection	1,950 rollouts	Exactly 50 rollouts for each of 39 tasks; 1,106 successes and 844 failures; all artifacts and official loading validated.
Xiaomi pooled training export	1,950 rollouts	Selected from a 2,642-rollout source pool; 1,105 successes and 845 failures; 38 tasks used for training after one sparse-failure exclusion.

The report bundle itself contains 20,857 archive entries and passed SHA-256 verification. Incomplete and quarantined artifacts are retained as provenance rather than silently deleted.

5 π_0 experiments

5.1 Training and evaluator smokes

Independent and LSTM two-epoch evaluator smokes first established that the official SAFE loader, detector checkpoints, and offline evaluator agreed. A two-configuration 1,000-epoch grid pilot then confirmed that the apparently short wall time was expected: with roughly 100 rollouts and batch size 512, one epoch is only a small full-batch update. These runs were engineering gates, not performance claims.

An initial six-run all-five-seen fit, before leakage-free hyperparameter selection, produced MLP ROC 0.506 ± 0.009 and PRC 0.541 ± 0.019 , versus LSTM ROC 0.702 ± 0.031 and PRC 0.675 ± 0.025 . It motivated the full inner-CV experiment but was superseded by the fixed-split result below.

5.2 Official 810-fit grid

All 810 runs completed (270 configurations, three seeds). The selected MLP achieved seen validation ROC **0.799 ± 0.081** , while its unseen value was **0.453 ± 0.037** . The LSTM achieved seen **0.792 ± 0.069** and unseen **0.581 ± 0.080** . These are validation results from a small 100-rollout dataset, not a final held-out estimate.

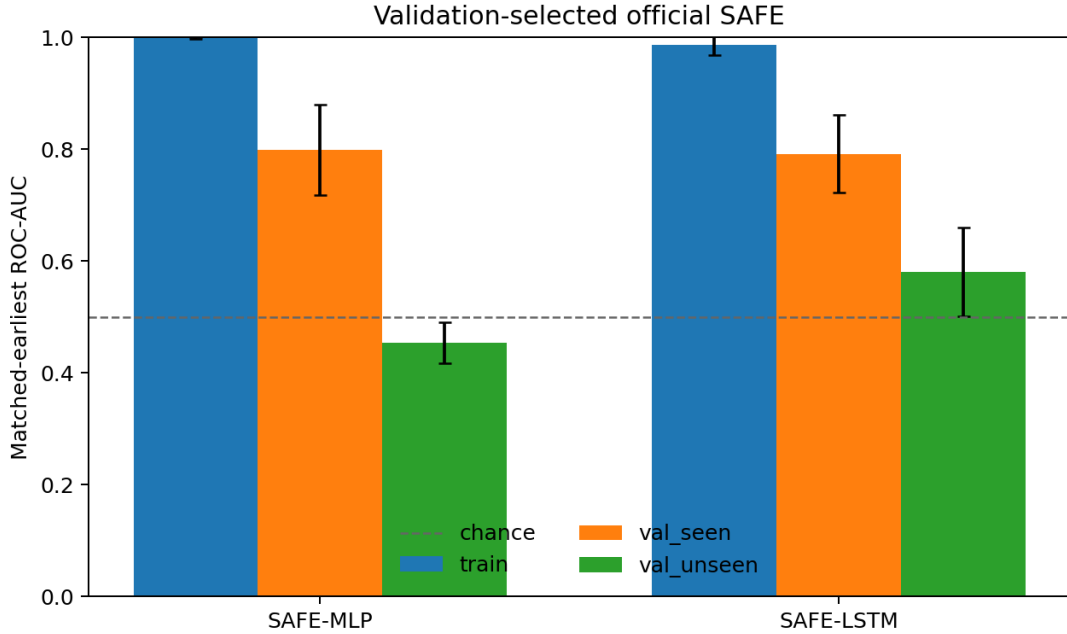


Figure 1: π_0 official-grid selection. Seen-task validation is substantially stronger than unseen-task validation, but uncertainty across seeds is high.

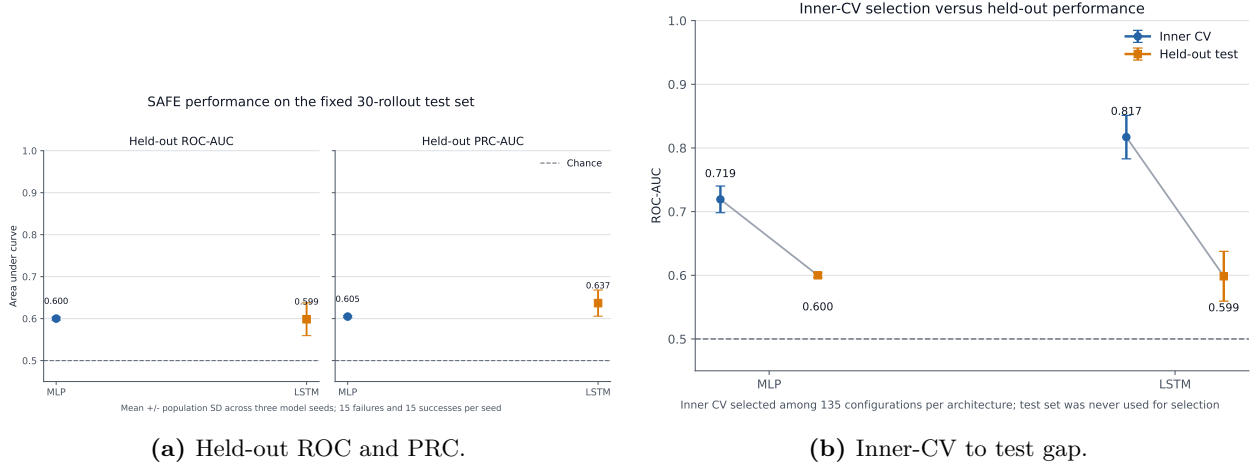
Conformal operation remained weak. At $\alpha = 0.15$, the MLP had TPR 0.300 and FPR 0.383; the LSTM had TPR 0.183 and FPR 0.150. Neither produced a convincing intervention point.

5.3 Leakage-free five-task experiment

The outer split used 70 training rollouts and 30 held-out rollouts, balanced within each of five atomic tasks. Inner-CV selected MLP ROC **0.719 ± 0.021** and LSTM ROC **0.817 ± 0.034** . Final held-out results were:

Table 2: π_0 leakage-free final results (three model seeds).

Model	Test ROC-AUC	Test PRC-AUC
Independent MLP	0.600 ± 0.004	0.605 ± 0.002
LSTM	0.599 ± 0.039	0.637 ± 0.031

**Figure 2:** Leakage-free π_0 evaluation. Stronger inner validation did not transfer to the small held-out set.

6 RLDX rollout-level experiments

6.1 Scaling from 10/10 to 25/25 per task

Ten tasks were used: five atomic and five composite. Increasing from 20 to 50 rollouts per task improved the independent MLP’s within-task ranking and reduced variance.

Table 3: RLDX held-out discrimination as data increased.

Dataset	Model	Raw pooled ROC	Train-only task-z ROC
10/10 per task	MLP	0.543 ± 0.007	0.674 ± 0.023
10/10 per task	LSTM	0.520 ± 0.025	0.537 ± 0.026
25/25 per task	MLP	0.575 ± 0.004	0.713 ± 0.017
25/25 per task	LSTM	0.554 ± 0.006	0.547 ± 0.010

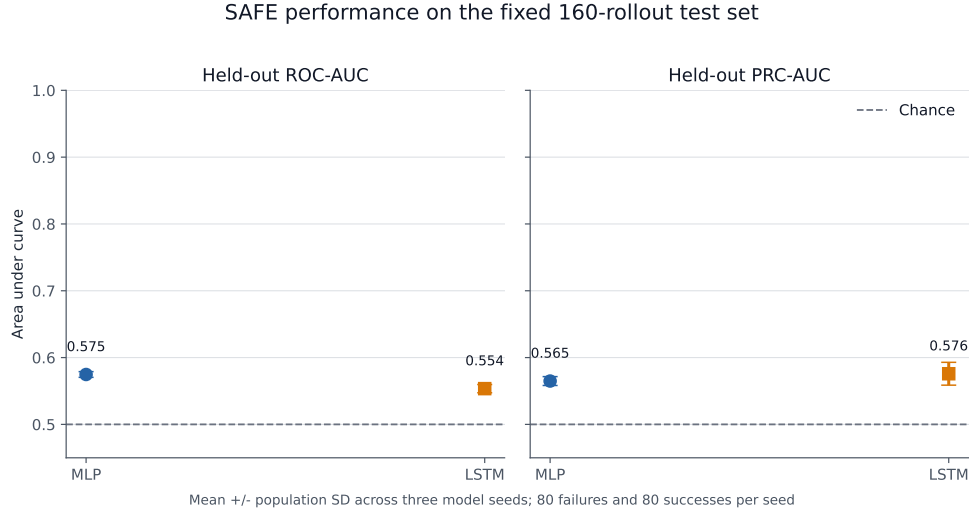


Figure 3: RLDX 25/25 final held-out metrics. The MLP is stable across seeds; the LSTM remains weaker.

The large raw-to-task-z gap means a global score is not directly comparable across tasks. Task normalization is reasonable as an analysis and calibration tool when task identity is known, but it is not evidence that the raw detector is globally calibrated.

6.2 Rollout-level conformal calibration

The 500-rollout dataset was divided into 340 training rollouts, 30 calibration successes, and 130 evaluation rollouts. At $\alpha = 0.15$, the MLP achieved TPR 0.196, FPR 0.040, and balanced accuracy 0.578. The originally archived detection-time value 0.716 was conditional on a failure being detected. Under the paper definition, an undetected failed rollout has normalized detection time 1, giving a paper-equivalent average of approximately 0.944. Recall therefore remained too low for reliable recovery.

6.3 Expanded alpha sweep and paper-equivalent timeliness

The functional conformal sweep was extended from $\alpha \leq 0.20$ to $\alpha \leq 0.50$. Increasing α lowers conservativeness, usually increasing both detection coverage and false alarms. Because the official `tfunc` modulation itself depends on α , the resulting functional bands are not guaranteed to be nested; small non-monotonic changes are therefore possible.

Table 4: Task-normalized RLDX MLP operating points. Accuracy is ordinary accuracy; balanced accuracy is the primary metric because each type-specific evaluation set contains 40 failures and 25 successes per seed.

Type	α	Accuracy	Balanced accuracy	TPR	FPR
Atomic	0.15	0.533 ± 0.015	0.608 ± 0.015	0.283	0.067
Atomic	0.20	0.615 ± 0.038	0.667 ± 0.030	0.442	0.107
Atomic	0.35	0.703 ± 0.032	0.683 ± 0.015	0.767	0.400
Atomic	0.50	0.708 ± 0.045	0.640 ± 0.048	0.933	0.653
Composite	0.15	0.446 ± 0.013	0.547 ± 0.007	0.108	0.013
Composite	0.20	0.508 ± 0.022	0.590 ± 0.021	0.233	0.053
Composite	0.35	0.487 ± 0.071	0.551 ± 0.053	0.275	0.173
Composite	0.50	0.554 ± 0.044	0.590 ± 0.032	0.433	0.253

At $\alpha = 0.20$, the combined ten-task detector had TPR 0.337, FPR 0.080, and balanced accuracy 0.629. Atomic $\alpha = 0.35$ is the best point under the paper’s two plotted objectives, but its 40% false-positive rate is unsuitable for routine intervention. Composite $\alpha = 0.50$ attains the same balanced accuracy as $\alpha = 0.20$ with nearly five times the FPR. The practical shared operating point remains $\alpha = 0.20$; the extended curve is a post-hoc sensitivity analysis, not a new confirmatory selection.

The SAFE paper defines T_{det} as the first threshold-crossing timestep divided by the failed rollout duration, assigns $T_{\text{det}} = 1$ to misses, and averages over ground-truth failed rollouts [1]. Figure 4 reconstructs that metric from the archived confusion counts and conditional detection times. It reproduces the paper’s evaluation geometry, not its complete benchmark: no human failure-onset annotations or competing baselines are available here.

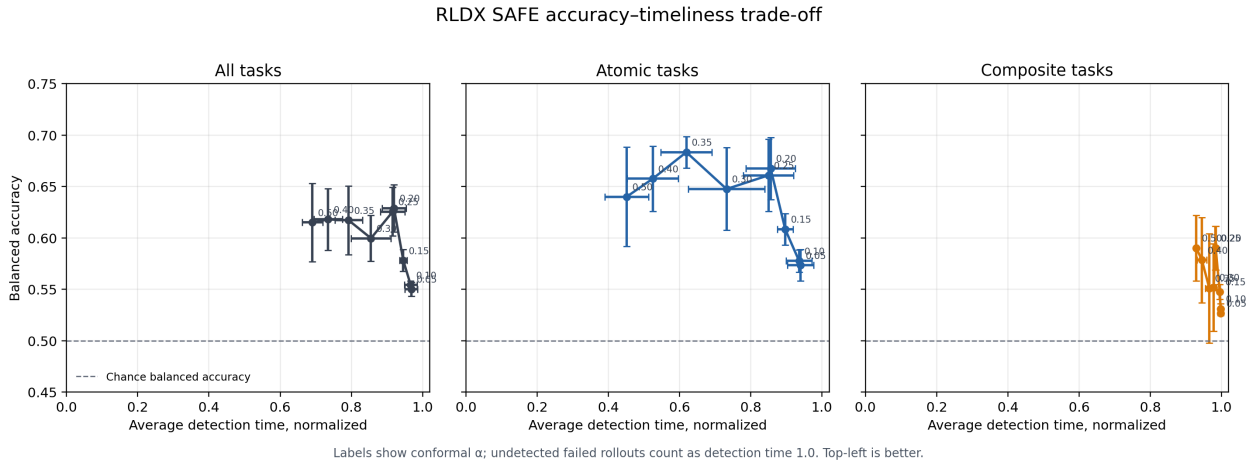


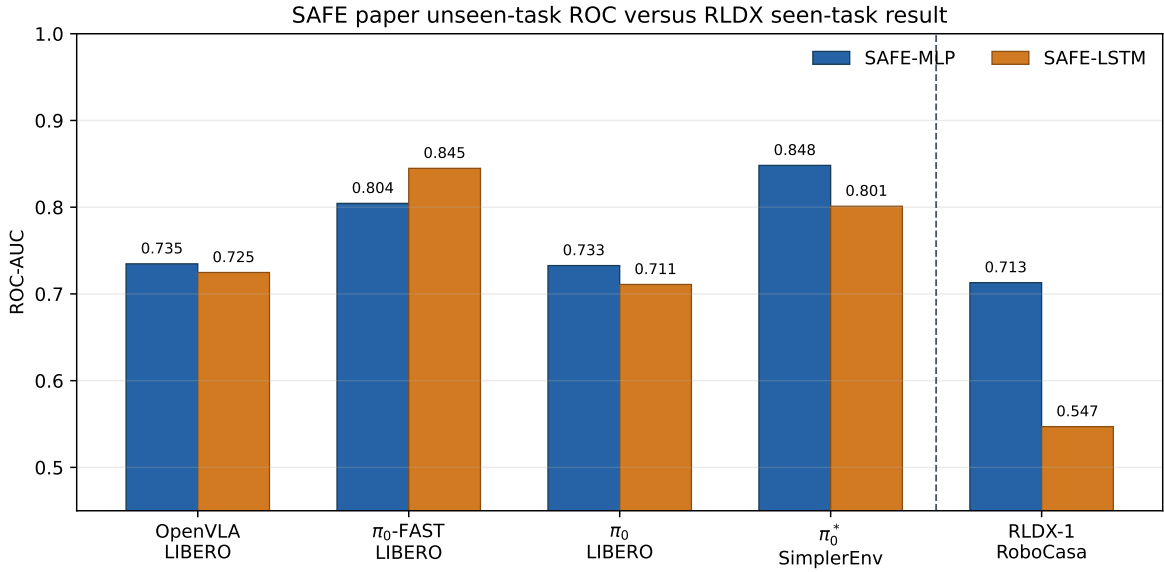
Figure 4: Paper-style balanced-accuracy–timeliness curves for task-normalized RLDX SAFE-MLP. Labels are conformal α ; misses count as normalized time 1. Atomic tasks form a useful frontier, whereas composite failures are generally detected near the horizon or missed. Error bars are population standard deviations across three model seeds on one fixed evaluation set.

6.4 Comparison with the SAFE paper

The original SAFE simulation study evaluated unseen tasks and reported ROC-AUC over the rollout maximum score. Its unseen-task SAFE-MLP ROC ranged from 0.733 on π_0 +LIBERO to 0.848 on π_0^* +SimplerEnv; SAFE-LSTM ranged from 0.711 to 0.845 [1]. The RLDX result uses held-out samples from the same ten tasks and training-only task-z normalization, so it is a seen-task conditional result rather than a zero-shot reproduction.

Table 5: SAFE paper unseen-task ROC-AUC and the present RLDX seen-task result. The protocols, policies, tasks, feature layers, and normalization differ; the last row is contextual, not a controlled ranking.

Policy and benchmark	SAFE-MLP	SAFE-LSTM
OpenVLA + LIBERO, unseen	0.735	0.725
π_0 -FAST + LIBERO, unseen	0.804	0.845
π_0 + LIBERO, unseen	0.733	0.711
π_0^* + SimplerEnv, unseen	0.848	0.801
RLDX-1 + RoboCasa, seen task-z	0.713	0.547
RLDX-1 + RoboCasa, seen raw pooled	0.575	0.554



Paper bars: unseen-task ROC. RLDX bars: train-only task-z ROC on held-out samples from the same ten tasks; values are not directly comparable.

Figure 5: Contextual ROC comparison. The dashed separator emphasizes that the SAFE-paper bars are unseen-task results, whereas RLDX uses held-out samples from known tasks and task-conditioned normalization.

The RLDX MLP task-z ROC 0.713 is numerically close to the paper’s π_0 +LIBERO unseen MLP ROC 0.733, but the similarity must not be interpreted as reproduced generalization. Raw RLDX pooling is materially weaker, the LSTM does not transfer, and the conformal composite-task curve lies near $T_{\text{det}} = 1$. Relative to the paper, the principal deficit is therefore not only retrospective ranking but the conversion of scores into timely, task-robust failure alarms.

6.5 Atomic versus composite ablation

The mixed MLP was stronger on atomic tasks (task-z ROC 0.827) than composites (0.667). Separate models did not improve matched evaluation:

Table 6: Matched mixed-versus-type-specific MLPs at $\alpha = 0.15$.

Type	Model	Task-z ROC	Task-z PRC	TPR	FPR	Bal. acc.
Atomic	Mixed	0.827	0.891	0.275	0.027	0.624
Atomic	Split	0.821	0.880	0.217	0.133	0.542
Composite	Mixed	0.667	0.773	0.167	0.067	0.550
Composite	Split	0.668	0.777	0.192	0.107	0.543

This rejects the hypothesis that mixing short- and long-horizon tasks is the main source of confusion. It instead motivates finer temporal labels for composites.

7 Natural prevalence and paper-style weighting

The natural-rate study compared matched balanced weighting, natural unweighted training, and natural training with official inverse-frequency weighting. Five subset seeds and three model seeds produced 90 completed runs on one immutable 160-rollout test set.

Table 7: Natural-prevalence study. AP denotes average precision.

Model	Regime	Raw ROC	Task-z ROC	Macro ROC	Raw AP	Task-z AP	Macro AP
MLP	Matched weighted	0.566	0.691	0.694	0.572	0.713	0.743
MLP	Natural unweighted	0.573	0.700	0.714	0.568	0.721	0.748
MLP	Natural weighted	0.571	0.699	0.715	0.563	0.726	0.751
LSTM	Matched weighted	0.546	0.538	0.556	0.579	0.556	0.645
LSTM	Natural unweighted	0.525	0.522	0.553	0.564	0.532	0.647
LSTM	Natural weighted	0.532	0.529	0.554	0.569	0.531	0.645

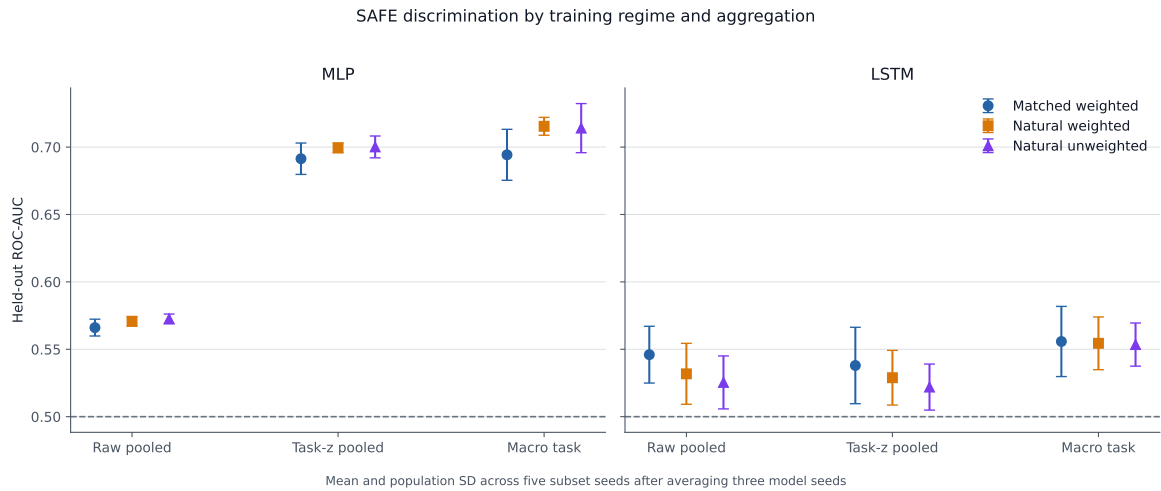


Figure 6: Aggregation view and sampling regime. Natural prevalence and weighting effects are small compared with the difference between raw and task-aware aggregation.

For the MLP, natural minus matched was $+0.008 \pm 0.012$ in task-z ROC and $+0.021 \pm 0.013$ in macro task ROC. Official weighting minus unweighted natural training was -0.001 ± 0.007 and $+0.001 \pm 0.013$. The finite pool is only mildly imbalanced globally, so this does not contradict the paper’s weighting choice; it shows that balancing was not the limiting factor here.

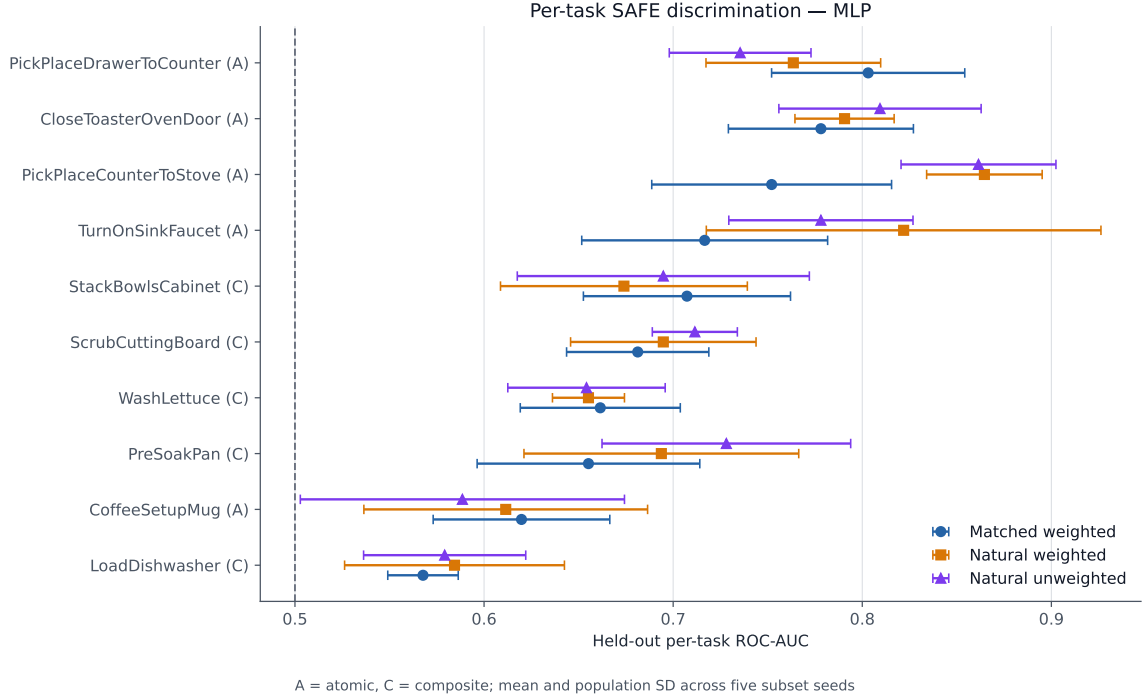


Figure 7: Per-task MLP ROC under the three sampling/weighting regimes. Effects differ by task but do not yield a consistent global advantage.

8 Semantic Subtask-SAFE

8.1 Target and decomposition

The active-stage target is “does the active natural-language subtask eventually fail before completion?” A successful parent contributes its completed subtask segments. A failed parent contributes its successful prefix plus exactly one failed active segment; unattempted future stages are excluded.

Predicate-only stages were not sufficient: natural-language tasks can combine several predicates, and adjacent actions may share a predicate. The final schema uses ordered task instructions, merges repeated-predicate stages, drops reset-satisfied stages, and preserves merge provenance.

8.2 Dataset and grouped split

Five composite tasks yielded 150 parents and 426 usable segments across 18 task/subtask identities. Failure support is sparse and skewed.

Table 8: Subtask-SAFE data and outer split.

Split	Parents	Segments	Success segments	Failure segments
Training	100	292	236	56
Held-out test	50	134	105	29
Total	150	426	341	85

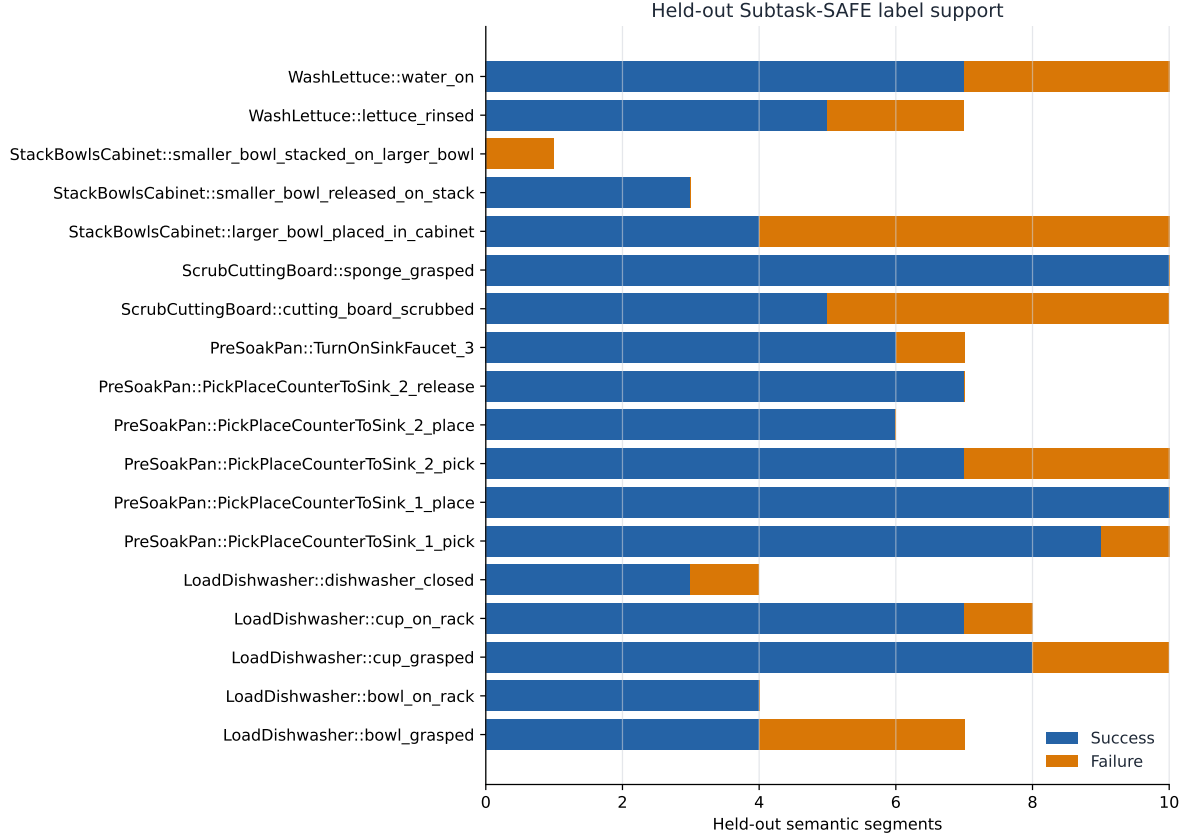


Figure 8: Support by semantic task/subtask identity. Early successful-prefix stages dominate; several stages have little or no failure support.

8.3 Inner CV and retrospective final result

All 810 inner-CV fits completed. MLP selection was horizon `concat-2`, diffusion `concat-2`, learning rate 10^{-3} , regularization 10^{-3} , with inner ROC 0.749 ± 0.037 . LSTM inner ROC was 0.722 ± 0.009 . All six final refits completed with official inverse-frequency class weights $[5.1228, 1.2321]$.

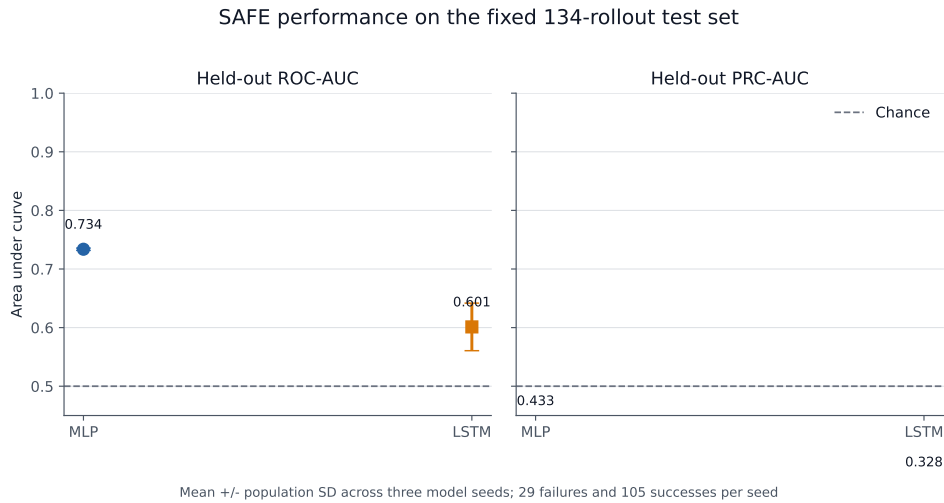


Figure 9: Retrospective held-out segment discrimination. MLP raw ROC is 0.734, but this uses the full active segment and pools heterogeneous stages.

The final retrospective results were MLP ROC 0.734 ± 0.002 , PRC 0.433 ± 0.012 ; LSTM ROC 0.601 ± 0.041 , PRC 0.328 ± 0.056 . These values motivated, but do not replace, causal evaluation.

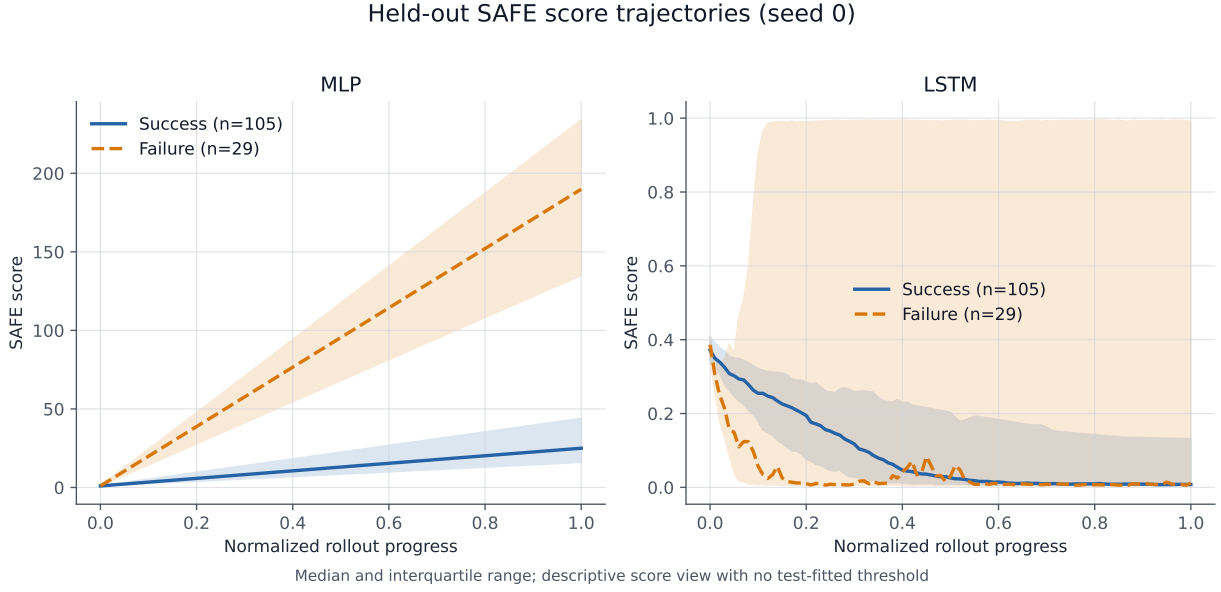


Figure 10: Example retrospective score trajectories. The detector often separates classes only after substantial within-stage evidence has accumulated.

9 Parent-aware causal analysis

9.1 Normalization and hierarchical uncertainty

Raw pooled MLP ROC was 0.734. Training-only task/subtask-z pooling reduced it to 0.561 ± 0.025 . A parent-rollout bootstrap gave mean 0.558, 95% CI [0.428, 0.676]. The elapsed-time baseline achieved mean 0.740, 95% CI [0.646, 0.827].

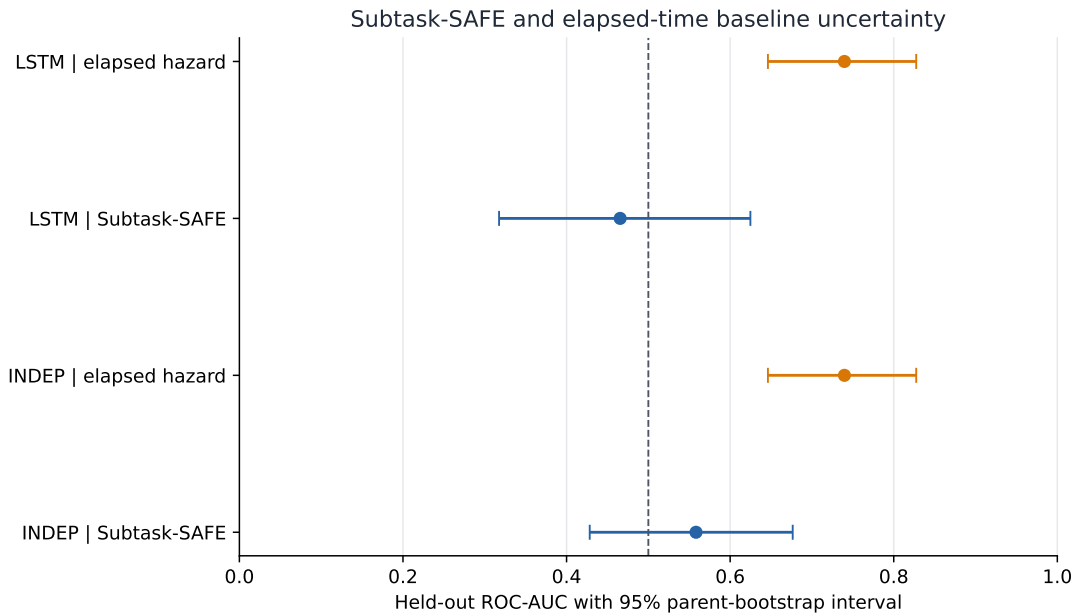


Figure 11: Hierarchical parent-rollout bootstrap. SAFE uncertainty overlaps chance, while elapsed time remains consistently stronger.

9.2 Causal prefix test

Scores were recomputed using only the first k true policy inferences of each active segment. This prevents future within-segment evidence from leaking into an early-warning claim.

Table 9: Causal-prefix ROC. Counts decrease because shorter segments are ineligible at longer prefixes.

k	Active segments	MLP SAFE	LSTM SAFE	Elapsed time
1	134	0.389	0.340	0.742
2	127	0.347	0.363	0.723
4	120	0.330	0.380	0.704
8	102	0.322	0.397	0.685
16	68	0.374	0.463	0.668
32	25	0.426	0.478	0.614

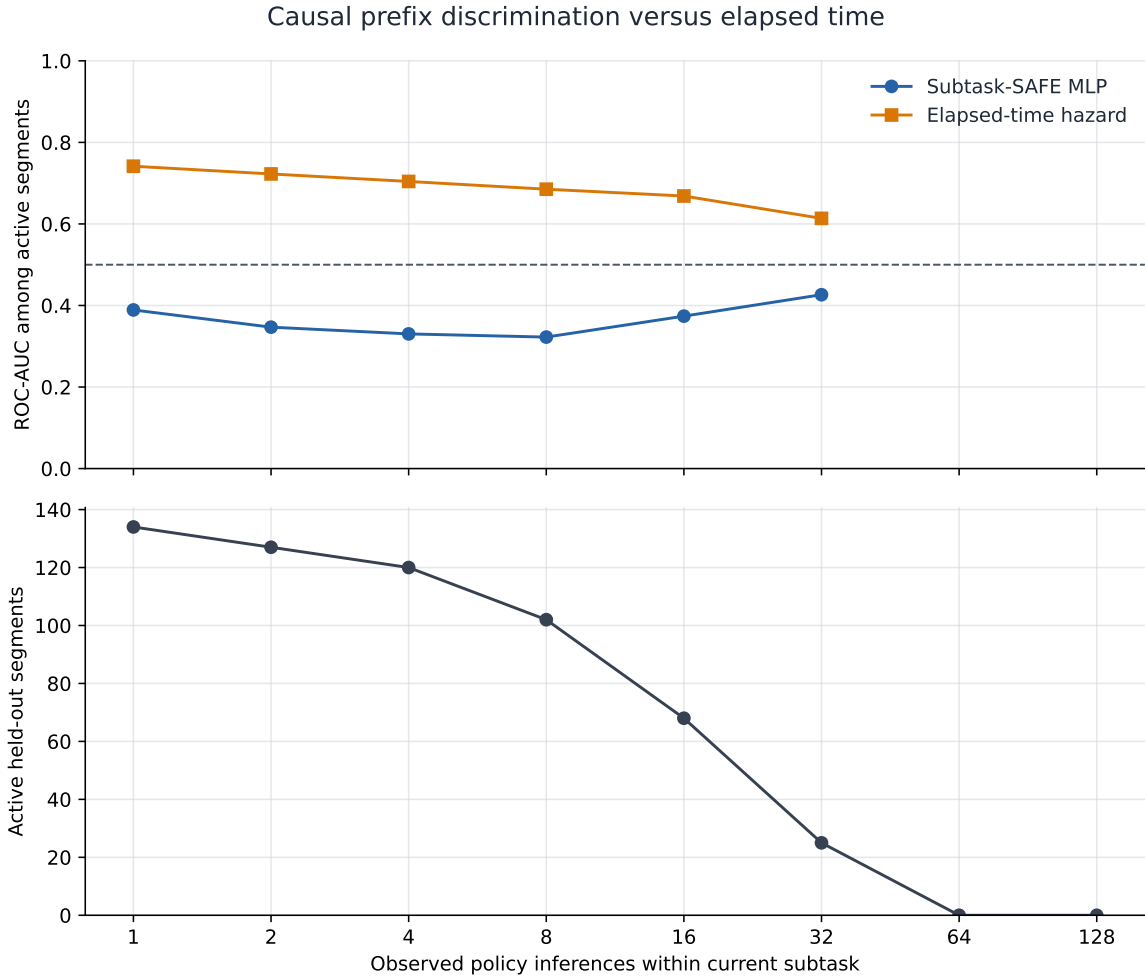


Figure 12: Causal-prefix analysis. Current SAFE scores are below chance at early prefixes; elapsed time is the dominant baseline at every supported prefix.

9.3 Parent-disjoint conformal operation

Calibration used 15 parents and evaluation used 35 disjoint parents. Reference and nonconformity calibration parents were disjoint. At $\alpha = 0.15$, SAFE achieved TPR 0.121, FPR 0.119, balanced

accuracy 0.501. Elapsed time achieved TPR 0.500, FPR 0.149, balanced accuracy 0.675.

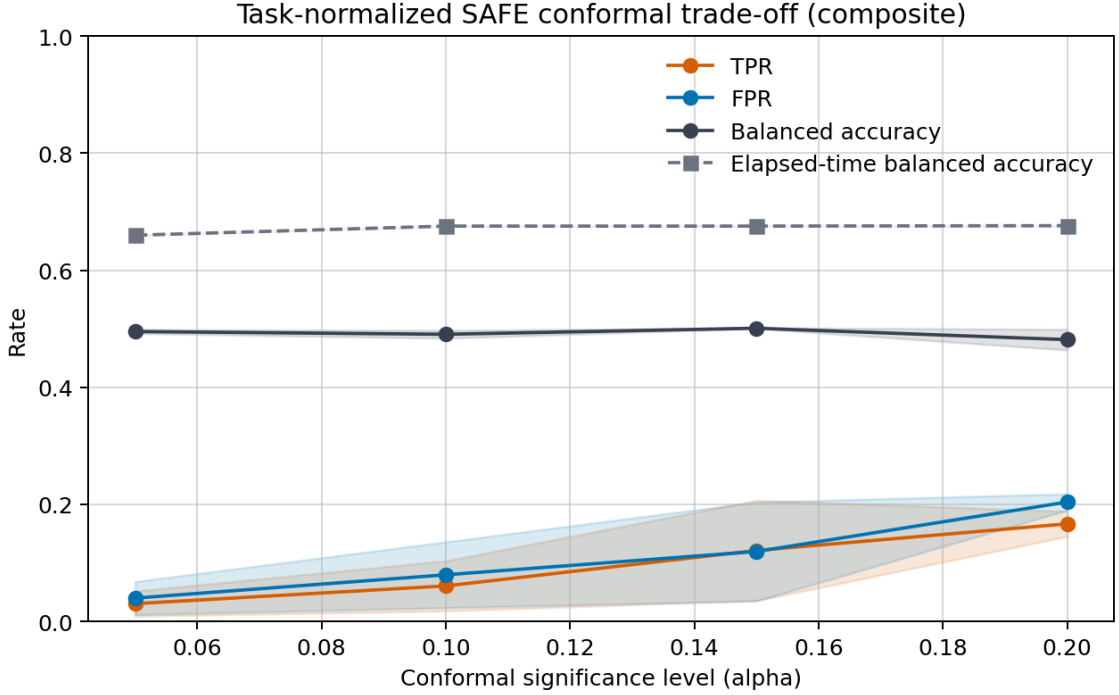


Figure 13: Parent-disjoint conformal trade-off. SAFE remains near chance across α ; elapsed time yields materially better balanced accuracy.

The reported SAFE lead of 1,661 environment steps at $\alpha = 0.15$ applies only to the 12.1% of failures detected. Lead time without detection coverage would be misleading.

10 Subtask-SAFE V2: causal objective and frozen evaluation

10.1 Continuation gates and four-stage pilot

The first causal analysis showed that full-segment discrimination was not a sufficient continuation criterion. V2 therefore introduced four explicit gates: causal-prefix performance, comparison with stage-conditioned elapsed time, parent-bootstrap uncertainty, and an operating point with useful TPR, controlled FPR, and positive lead. The data plan initially selected four stages with both scientific value and some failure support:

- LoadDishwasher::cup_on_rack;
- PreSoakPan::TurnOnSinkFaucet_3;
- ScrubCuttingBoard::cutting_board_scrubbed;
- WashLettuce::water_on.

Their combined segment deficit was 127. Stage-only conditioning gave mean fold ROC approximately 0.599, while stage plus elapsed progress gave approximately 0.588. A broader policy-only/random-prefix ablation selected horizon concat-2, diffusion selector 1.0, learning rate 3×10^{-4} , and regularization 0.01, with inner-validation ROC 0.645 ± 0.040 .

The first gated refit did not justify more undirected collection. Its causal-prefix ROC was 0.575 versus 0.673 for elapsed time; TPR was 0.854 only because FPR rose to 0.851. The parent-bootstrap interval for SAFE minus elapsed ROC was $[-0.240, 0.037]$. This result motivated a change from the terminal segment label to a within-horizon event target.

10.2 Within-horizon labels and representation screen

For a prefix ending at inference t , the V2 label asks whether the active stage fails within the next H policy inferences. A feasibility scan showed that $H = 128$ with evaluation prefix 32 was the most robust candidate: all three inner folds retained both classes. By contrast, $H = 96$ at prefix 48 produced fold ROCs 0.717, 0.250, and 0.400, indicating unstable support.

Fifteen representation/objective treatments were compared on the feasible pilot. Table 10 lists the highest-ranking choices. The official SAFE loss on raw features ranked first; temporal summaries were competitive but not uniformly better.

Table 10: Top V2 representation/loss treatments in the small within-horizon screen. These are inner-validation results, not frozen-test estimates.

Representation and loss	Mean ROC	SD
Raw, official SAFE	0.914	0.053
Raw + delta + mean + slope, official SAFE	0.867	0.066
Raw + delta + mean + slope, BCE	0.838	0.046
Raw + delta + mean + slope, focal	0.822	0.053
Raw + delta, official SAFE	0.798	0.087

The corresponding small final pilot appeared promising: raw features reached ROC 0.819 and temporal summaries 0.956 at prefix 32. However, stage support was only 1/1, 4/0, 3/5, and 7/0 success/failure examples. Each seed also had only five calibration successes, giving nominal conformal resolution $1/(5+1) = 0.167$. At target FPRs up to 0.30 the temporal detector produced zero TPR; relaxing the target to 0.40 yielded realized FPR 0.111 and TPR 0.833. The high ROC was therefore treated as a collection pilot, not as a result.

10.3 Target-aware collection and immutable allocation

Targeted collection added 400 new-seed parent rollouts. The initial 80-parent batch exposed that the chosen intermediate stages did not consistently accumulate failures. The viable target set was consequently fixed to:

- `LoadDishwasher::dishwasher_closed;`
- `PreSoakPan::TurnOnSinkFaucet_3;`
- `ScrubCuttingBoard::cutting_board_scrubbed;`
- `WashLettuce::lettuce_rinsed.`

The long collection contributed 120 LoadDishwasher, 120 PreSoakPan, 20 ScrubCuttingBoard, and 30 WashLettuce parents; an additional 30 WashLettuce parents closed the final deficit. Every source dataset passed the rollout, feature, video, and semantic-segment validators.

Before scoring, an immutable target-aware allocation selected five successful calibration parents and 10 successful plus 10 failed evaluation parents per stage. Thus calibration contains 20 parents and evaluation contains 80, with no identity overlap. The remaining 180 eligible new-seed parents were later added to training, while the frozen 100-parent allocation remained byte-for-byte unchanged. The final training manifest contains 280 parents, of which 244 reach a selected target stage; the original old test parents were excluded.

10.4 Frozen-holdout result

On the expanded training set, raw inner CV selected horizon `concat-2`, diffusion 1.0, learning rate 3×10^{-5} , and regularization 0.001, with ROC 0.712 ± 0.078 . Temporal summaries selected horizon 1.0, diffusion 1.0, learning rate 10^{-4} , and regularization 0.001, with ROC 0.703 ± 0.029 .

Evaluation at the predeclared causal prefix 32 reversed the optimistic pilot. Raw features achieved ROC 0.512 ± 0.003 ; raw plus delta, mean, and slope achieved 0.566 ± 0.004 . Figure 14 shows the stage-level heterogeneity.

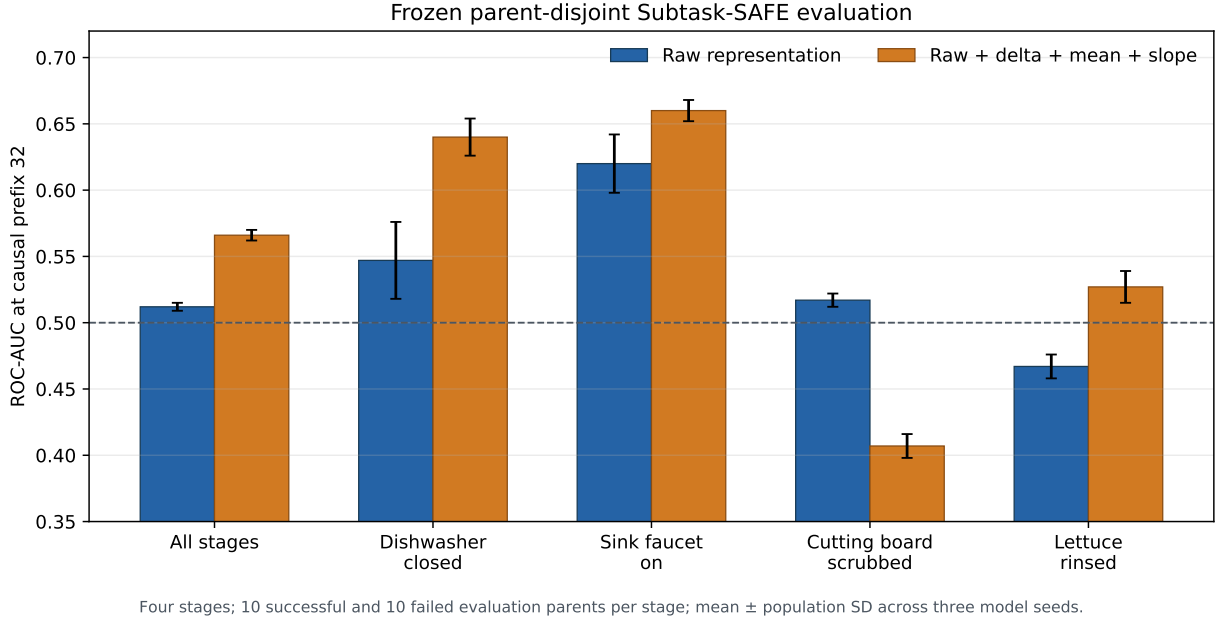


Figure 14: Final V2 result on the frozen 80-parent evaluation allocation at causal prefix 32. Each stage contains 10 successful and 10 failed parents. Temporal summaries help on dishwasher closure and faucet activation, hurt cutting-board scrubbing, and remain inconclusive overall.

Table 11: Frozen prefix-32 ROC-AUC, mean $\pm$ population SD across three model seeds.

Evaluation group	Raw	Raw + delta + mean + slope
All four stages	0.512 ± 0.003	0.566 ± 0.004
Dishwasher closed	0.547 ± 0.029	0.640 ± 0.014
Sink faucet on	0.620 ± 0.022	0.660 ± 0.008
Cutting board scrubbed	0.517 ± 0.005	0.407 ± 0.009
Lettuce rinsed	0.467 ± 0.009	0.527 ± 0.012

All-prefix segment ROC was 0.733 for raw features and 0.738 for temporal summaries, but those values pool 675 prefix examples from multiple times and are not the early-warning endpoint. At prefix 32, raw operation gave TPR 0.217, FPR 0.083, and normalized lead 0.755; temporal operation gave TPR 0.133, FPR 0.142, and lead 0.779. Parent-bootstrap confidence intervals for SAFE ROC included chance, and SAFE-minus-duration intervals included zero. Both treatments therefore failed the continuation gates.

This is the most important negative result in the report: additional target-aware data greatly

improved class support but did not produce task-robust early discrimination. The large pilot-to-frozen gap shows why target identities and allocation must be frozen before representation selection.

11 Xiaomi-Robotics-1 original SAFE

11.1 Scope and initial ten-task study

Every Xiaomi result uses only the original binary terminal task label. No Subtask-SAFE, progress predicate, recovery annotation, or failure-onset label is present. The initial set contained:

- five atomic tasks: CoffeeSetupMug, TurnOnMicrowave, TurnOffStove, CloseBlenderLid, and SlideDishwasherRack;
- five composites: DeliverStraw, ArrangeTea, WashFruitColander, GarnishPancake, and WaffleReheat.

The first 500 natural rollouts contained 266 successes and 234 failures. Additional attempts were retained until each task had at least 25 of both outcomes, producing a 692-rollout pool with 419 successes and 273 failures. A deterministic seed-0 selection yielded an official-loader-compatible 500-rollout export with exactly 25 successes and 25 failures per task and 40,160 policy records.

Table 12: Initial Xiaomi 10-task selection and final held-out results. Hyperparameters were selected without using the 160-rollout outer test.

Model	Selected (h, d, lr, λ)	Inner ROC	Test ROC	Test AP	Duration ROC
MLP	$(1, 1, 10^{-4}, 10^{-3})$	0.534 ± 0.008	0.517 ± 0.001	—	0.743
LSTM	$(c2, 1, 10^{-3}, 10^{-2})$	0.595 ± 0.029	0.550 ± 0.024	—	0.743

At fixed conformal $\alpha = 0.15$, normalized MLP and LSTM balanced accuracy was 0.499 ± 0.014 and 0.522 ± 0.038 , respectively. Matched-success-length and fixed-25%-landmark controls removed most duration discrimination but did not produce a strong detector: their best outer ROC values were 0.546 for MLP and 0.530 for LSTM. A causal-prefix experiment reached MLP task-macro ROC 0.610 at 10% and 0.608 at 25%, but a staged cascade beat time-only on only two of 240 seed-level failed predictions. This motivated broader collection.

11.2 Official mid-success task collection

The broad task set was frozen from Xiaomi’s released 2,500-rollout RoboCasa365 summary. The inclusive selection rule was 5 through 45 successes out of 50. It selected 39 tasks; tasks at 0/50 or 50/50 were excluded. Two resumable GPU shards collected exactly 50 new natural rollouts per task with base seed 100007. All successes, failures, quota overshoot, and historical errors were preserved.

Table 13: Official Xiaomi success count and the new 50-rollout collection.

Task	Type	Official	Fresh success	Fresh failure
CategorizeCondiments	composite	5	7	43
PrepareCoffee	composite	8	9	41
PortionHotDogs	composite	10	8	42
SearingMeat	composite	13	15	35
PackIdenticalLunches	composite	14	14	36
WeighIngredients	composite	14	15	35

Task	Type	Official	Fresh success	Fresh failure
BreadSelection	composite	15	17	33
CuttingToolSelection	composite	17	22	28
RecycleBottlesByType	composite	17	9	41
CloseBlenderLid	atomic	18	17	33
GetToastedBread	composite	18	20	30
MakeIceLemonade	composite	19	19	31
StirVegetables	composite	19	24	26
WaffleReheat	composite	21	29	21
ArrangeTea	composite	23	27	23
DeliverStraw	composite	23	18	32
CoffeeSetupMug	atomic	25	20	30
WashFruitColander	composite	27	31	19
TurnOnMicrowave	atomic	28	33	17
GarnishPancake	composite	29	31	19
ArrangeBreadBasket	composite	30	25	25
RinseSinkBasin	composite	30	35	15
TurnOffStove	atomic	30	30	20
SteamInMicrowave	composite	32	35	15
LoadDishwasher	composite	36	36	14
SetUpCuttingStation	composite	36	32	18
StackBowlsCabinet	composite	36	40	10
KettleBoiling	composite	37	29	21
PreSoakPan	composite	38	38	12
ScrubCuttingBoard	composite	40	35	15
SlideDishwasherRack	atomic	40	42	8
StoreLeftoversInBowl	composite	42	43	7
TurnOnSinkFaucet	atomic	42	38	12
WashLettuce	composite	42	36	14
PickPlaceCounterToCabinet	atomic	43	44	6
PickPlaceSinkToCounter	atomic	44	49	1
CloseToasterOvenDoor	atomic	45	44	6
OpenCabinet	atomic	45	46	4
PickPlaceCounterToStove	atomic	45	44	6
Total	39 tasks	—	1,106	844

Deep validation found 1,950 unique rollout IDs and task/seed/reset identities, no missing required artifacts, and no loader incompatibility. Twelve historical error records document recovered attempts. The two shards occupied about 36 GiB.

11.3 Pooling, split, and detector selection

The 1,950 new rollouts were merged with the earlier 692, giving an immutable 2,642-rollout pool. Deterministic within-task/outcome selection retained exactly 50 per task: 1,105 successes and 845 failures. The `PickPlaceSinkToCounter` group had only one failure and was excluded from model development, leaving 38 tasks.

Table 14: Frozen broad Xiaomi split and hyperparameter search.

Property	Value
Outer training	1,824 (1,018 success; 806 failure)
Untouched outer test	76 (38 success; 38 failure)
Held-out support	1 success and 1 failure per task
Grid	$3h \times 3d \times 5$ learning rates $\times 3$ regularizers
Inner folds/model seeds	3 / 3
Completed fits	810/810; zero failures

The MLP selected ($h = 1, d = 1, \text{lr} = 10^{-4}, \lambda = 10^{-3}$) with inner ROC 0.654 ± 0.025 . The LSTM selected ($h = \text{concat-2}, d = \text{concat-2}, \text{lr} = 3 \times 10^{-4}, \lambda = 0.1$) with inner ROC 0.742 ± 0.012 . All six final refits completed on identical identities.

Table 15: Broad Xiaomi final refits and training-only task normalization.

Model	Raw ROC	Task-z ROC	Raw AP	Task-z AP	Macro task ROC
MLP	0.532 ± 0.016	0.692 ± 0.013	0.528	0.733	0.689
LSTM	0.628 ± 0.008	0.646 ± 0.017	0.640	0.663	0.640

Completed-duration ROC was 0.804. Normalization reversed the raw architecture ordering: the MLP has the best within-task ranking, whereas the LSTM benefits more from between-task score offsets.

11.4 Atomic/composite and causal online replay

The 38-task set contains ten atomic and 28 composite tasks. Because each task has only one held-out example per class, the task-type estimates are imprecise.

Table 16: Atomic versus composite outer-test results.

Model	Type	Tasks	Raw ROC	Task-z ROC	Task-z AP	Macro ROC	Time ROC
MLP	Atomic	10	0.573 ± 0.045	0.730 ± 0.054	0.792 ± 0.053	0.600	1.000
MLP	Composite	28	0.539 ± 0.018	0.680 ± 0.025	0.729 ± 0.015	0.720 ± 0.017	0.912
LSTM	Atomic	10	0.627 ± 0.098	0.630 ± 0.123	0.727 ± 0.098	0.633 ± 0.125	1.000
LSTM	Composite	28	0.660 ± 0.027	0.651 ± 0.021	0.647 ± 0.019	0.643 ± 0.051	0.912

The MLP macro atomic-minus-composite difference was -0.120, with task-bootstrap 95% interval $[-0.416, 0.166]$; the LSTM difference was -0.010 with interval $[-0.264, 0.245]$. No reliable task-type difference is established.

Causal replay used only the current prefix, task identity, training-derived horizon, and active status. It split the 1,824 training rollouts into 1,748 meta-fit and 76 threshold-validation rollouts; the opened 76-rollout outer test was unchanged.

Table 17: Causal online event replay. Detection is fraction of task horizon; adjusted assigns missed failures 1.0.

Model	Detector	Event ROC	TPR	FPR	Balanced accuracy	Detection	Adjusted
MLP	Time only	0.965	0.842	0.053	0.895	0.786	0.820
MLP	SAFE only	0.951 ± 0.011	0.983 ± 0.025	0.132	0.925 ± 0.012	0.662 ± 0.021	0.668 ± 0.028
MLP	SAFE+time+task	0.968 ± 0.007	0.965 ± 0.025	0.053	0.956 ± 0.012	0.842 ± 0.013	0.848 ± 0.008
LSTM	SAFE only	0.814 ± 0.032	0.895 ± 0.057	0.377 ± 0.054	0.759 ± 0.033	0.346 ± 0.032	0.415 ± 0.049
LSTM	SAFE+time+task	0.863 ± 0.019	0.781 ± 0.090	0.272 ± 0.045	0.754 ± 0.022	0.435 ± 0.040	0.556 ± 0.074

At fixed causal landmarks, MLP SAFE task-macro ROC was 0.671 at 10% and 0.626 at 25%; LSTM values were 0.667 and 0.616. Task-conditioned time-only is exactly 0.5 within task at a fixed landmark. This is evidence of early internal signal, but not yet of a better operating policy: MLP SAFE obtains earlier detection by accepting five false alarms among 38 successes, versus two for time-only.

11.5 Frozen prospective matched-FPR experiment

The confirmatory next step is a two-phase shadow protocol. It preserves all three selected MLP seeds, fits per-seed normalization from the original 1,824 training rollouts, takes each seed’s running maximum, and averages the three normalized trajectories. A new calibration batch targets three successes and three failures per task (228 primary outcome-quota rollouts, plus retained overshoot). It selects global thresholds under one empirical event-level $\text{FPR} \leq 0.05$ constraint. The staged detector permits SAFE from rollout start through 25%, suppresses alarms from 25–50%, and enables the frozen task-conditioned time fallback from 50% onward. Its two thresholds are searched jointly under the same rollout-level FPR cap.

After the runtime bundle is frozen, a disjoint shadow test targets ten successes and ten failures per task (760 primary outcome-quota rollouts, plus retained overshoot). Frozen thresholds are replayed causally at every recorded inference, but never stop, retry, recover, or change the Xiaomi policy. Primary endpoints are Wilson-interval TPR/FPR, miss-adjusted detection fraction, recall by 10/25/50%, paired earlier/same/later/missed counts, task-macro landmark ROC, task/rollout bootstrap intervals, FPR drift, and atomic/composite strata.

The staged detector passes only if FPR remains compatible with 5%, TPR is no more than 0.05 below time-only, adjusted detection is at least 0.10 earlier, at least 25% of failures are detected by 25%, and the paired bootstrap favors earlier detection without material excess false alarms. Recovery is a separate randomized intervention experiment and is not estimated from shadow data.

12 Interpretation across all experiments

12.1 What the evidence supports

1. The feature-capture and official-export engineering is valid for both π_0 and RLDX.
2. RLDX MLP features contain useful within-task retrospective failure information.
3. Atomic tasks are materially easier than composite tasks.
4. Task and stage score offsets are large; task-aware normalization is reasonable when fitted without test leakage and when task identity is available.
5. Semantic decomposition is necessary for valid subtask labels and grouped evaluation.
6. Parent-aware causal-prefix testing is necessary before claiming early warning.
7. A frozen, stage-balanced evaluation allocation prevents promising but support-limited pilot results from becoming final claims.

8. Xiaomi’s normalized MLP contains early within-task causal signal across a much broader task set, making it the current best candidate for a matched-FPR shadow test.

12.2 What the evidence rejects

1. More complex temporal modeling alone: the LSTM was not competitive.
2. Mixed horizons alone as the cause: split atomic/composite models did not beat the mixed MLP.
3. Balanced sampling alone as the cause: natural prevalence and paper-style weighting had negligible paired effects.
4. Semantic decomposition alone as the solution: Subtask-SAFE raw ROC did not translate into early causal discrimination.
5. Within-horizon relabeling, stage conditioning, and temporal summaries as sufficient fixes: all improved selected pilots, but failed to transfer to the immutable four-stage test.
6. More of the same target-aware data as a guaranteed remedy: expanding to 280 training parents did not make prefix-32 ROC statistically distinguishable from chance.
7. Raw pooled Xiaomi scores as the deployment variable: training-only task normalization is necessary for the MLP, and elapsed time remains a stronger raw retrospective predictor.

12.3 Why normalization is reasonable, but insufficient

Training-only task or task/subtask normalization is scientifically reasonable because different tasks have different score scales and the deployed controller knows its commanded task. It is analogous to task-conditional calibration, not test-set tuning. However, normalization can only remove offsets; it cannot create within-stage ordering. The Subtask-SAFE drop from raw ROC 0.734 to parent-aware 0.558 exposes that limitation. Xiaomi is the complementary case: MLP task-z ROC rises from 0.532 to 0.692, indicating useful within-task ordering hidden by task offsets. Every policy family still requires its own training-derived calibration; thresholds and normalization are not transferable.

13 Recommended next experiments

1. **Run the Xiaomi prospective matched-FPR shadow test first.** Use the already selected three-seed MLP ensemble, a fresh 3/3-per-task calibration batch, a disjoint 10/10-per-task test, a jointly constrained 5% FPR cascade, and no policy intervention.
2. **Freeze before test collection.** Persist checkpoint and config checksums, training-only normalization, task horizons, time-risk curves, thresholds, stage windows, task list, and calibration identities. Never retune these values after test outcomes are visible.
3. **Retain every attempt.** Outcome quotas define the primary balanced endpoint, while the complete overshoot stream supplies an all-retained sensitivity analysis and documents collection cost and errors. Because stopping depends on both class quotas, this stream is not an unbiased deployment-prevalence sample.
4. **Keep paired timeliness primary.** Compare staged SAFE and time-only on identical failed rollouts with misses assigned fraction 1.0; report recall by 10/25/50%, Wilson intervals, and task/rollout bootstrap differences.
5. **Gate recovery on the frozen success criteria.** Only a prospective pass permits a separately powered randomized shadow-versus-recovery intervention study.
6. **Continue RLDX diagnosis separately.** Stage-specific heads, survival objectives, earlier feature layers, and complementary proprioceptive/contact signals remain appropriate RLDX follow-ups, but they must not be mixed with Xiaomi original-SAFE evidence.

14 Limitations

- π_0 and RLDX use different policies, feature shapes, tasks, and data volumes; absolute values are not direct architecture comparisons.
- Only three detector seeds were used; π_0 held-out sets were particularly small.
- Natural-rate subsets reflect the collected pool, not an unconstrained deployment distribution.
- Failure support is sparse for many semantic stages; segment classes are strongly imbalanced.
- Segments within one parent are dependent; only the parent-aware analyses model this hierarchy explicitly.
- The original causal-prefix study falls to 25 segments at 32 inferences and zero at 64; V2 fixes prefix-32 balance for four stages but does not test unseen task identities.
- The frozen 100-parent allocation was inspected in an earlier target-aware experiment before the augmented refit. It is a locked paired follow-up set, not a pristine new confirmatory test.
- Only four composite stages have balanced V2 evaluation support; per-stage confidence remains limited at 10 successes and 10 failures.
- RoboCasa predicate boundaries are privileged simulator information; a real robot requires a separate online subtask-progress estimator.
- Conformal validity depends on exchangeability and can fail under task, policy, layout, or seed drift.
- The broad Xiaomi outer test has only one success and one failure per task; task-level and atomic/composite uncertainty is therefore wide.
- Xiaomi causal replay is post-hoc because the 76 outer-test trajectories were already opened by raw and normalized analyses.
- The prospective Xiaomi calibration and test described here were not yet run at report time; their sample counts are minimum targets, not completed evidence.

A Experiment roots

Experiment	Cluster path
π_0 official grid	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_pi0_official_grid_2gpu_20260716_153526
π_0 leakage-free CV	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_all5_seen_innervcv_20260717_012538
RLDX 10-task CV	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_seen10_official_cv_2gpu_20260727_135354
RLDX 10-task final	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_seen10_final_refit_20260727_155824
RLDX 25/25 CV	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_25x25_cv_2gpu_20260729_203916
RLDX 25/25 final	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_25x25_final_2gpu_20260730_104400
RLDX calibration	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_25x25_final_2gpu_20260730_104400_calibration_20260730_113807
RLDX expanded sweep	alpha /gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_25x25_final_2gpu_20260730_104400_alpha_sweep_005_to_050_20260806_142630
Task-type ablation	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_25x25_type_ablation_20260730_115819
Natural-rate study	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/safe_rldx1_natural_rate_screen_20260730_161528

Subtask source data	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/rldx1_composite_subtask_seed8_9_10_150_20260803_210535
Subtask official export	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/rldx1_subtask_safe_official_20260803_212413
Subtask CV	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/rldx1_subtask_safe_cv_2gpu_20260803_222020
Subtask final/causal	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/rldx1_subtask_safe_final_2gpu_20260804_150002
Subtask V2 pilot	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/rldx1_subtask_v2_final_20260804_185021
Target-aware frozen pool	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/rldx1_target_aware_holdout400_20260805_162027
Augmented split manifest	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/rldx1_subtask_augmented280_frozen100_20260805_190520
Augmented raw/temporal CV	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/rldx1_augmented280_frozen100_cv_2gpu_20260805_205645
Augmented frozen final	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/rldx1_augmented280_frozen100_final_2gpu_20260806_132421
Xiaomi 10-task balanced export	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/xr1_safe_balanced_25x25_seed0_20260807_122434/official_safe_25x25_seed0
Xiaomi 39-task collection	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/xr1_safe_official_5to45_50each_20260808_215640
Xiaomi 2,642-rollout pool	/gs/bs/tga-shinoda/felid/robocasa_rollouts/safe/xr1_safe_pooled_39tasks_50each_seed0_20260812_221815/source_pool_all_2642
Xiaomi broad final refits	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/xr1_pooled_39tasks_final_refits_20260814_231508
Xiaomi causal replay	/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/xr1_pooled_39tasks_final_refits_20260814_231508/causal_online_replay_20260815_003121

B Reproducibility and figure map

The repository chronology is maintained in `docs/safe_robocasa_experiment_report.md`. Core entry points are:

- `collect_atomic_rollouts`, `collect_mixed_rollouts`, and `export_official_safe`;
- `run_seen_cv_grid` and `train_seen_tasks`;
- `score_seen_checkpoint_external` and `run_prospective_matched_fpr`;
- the parent-aware Subtask-SAFE analysis and calibration modules.

Most figures in this PDF are copied unchanged from the checksum-verified evidence bundle. The 6 August comparison and frozen-prefix figures are deterministic renderings of the report CSVs; their generator is `generate_update_figures.py`. `chart_map.md` records the source path and interpretation for every visual. Machine-readable supporting CSVs are in `tables/`. Videos remain in the evidence bundle; they are not embedded in PDF because their timing and score overlays are better inspected as MP4. The score-video tool now supports a full-rollout view and filenames that identify normalization, ground truth, and detector correctness.

C Metric-reading checklist

1. Is the metric retrospective full-segment or causal-prefix?
2. Is the score raw pooled, train-only normalized, or macro per task/stage?

3. Are dependent segments grouped by parent rollout?
4. Was the outer test excluded from hyperparameter selection and calibration fitting?
5. Is elapsed time evaluated on the same identities?
6. For conformal results, are TPR, FPR, balanced accuracy, and lead time reported together?

References

- [1] Gu, J., Kirmani, S., et al. SAFE: Multitask failure detection for vision-language-action models. *Advances in Neural Information Processing Systems*, 2025. [SAFE project page and paper](#).